

Compound Helicopter – Individual Assignment 2

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Section 1:

Additional Assumptions & Data

1.1 Individual Assumptions/Data

Airframe & Weight

- **Gross Weight:** 4050 kg
- **Fuselage Drag Area:** 1.45 m² (more aerodynamic)
- **Shaft Inclination:** -2.5° (optimized for high-speed flight)
- **Fuel Capacity:** 800 kg

Main Rotor System

- **Description:** 5-bladed main rotor
- **Airfoil:** SC1095
- **Rotor Radius:** 6.6 m
- **Rotor Speed:** 315 RPM
- **Chord Length:** 0.38 m (constant)
- **Twist Variation:** -12° (linear)
- **Root Cutout:** 20% of radius
- **Lock Number:** 7 (more responsive)

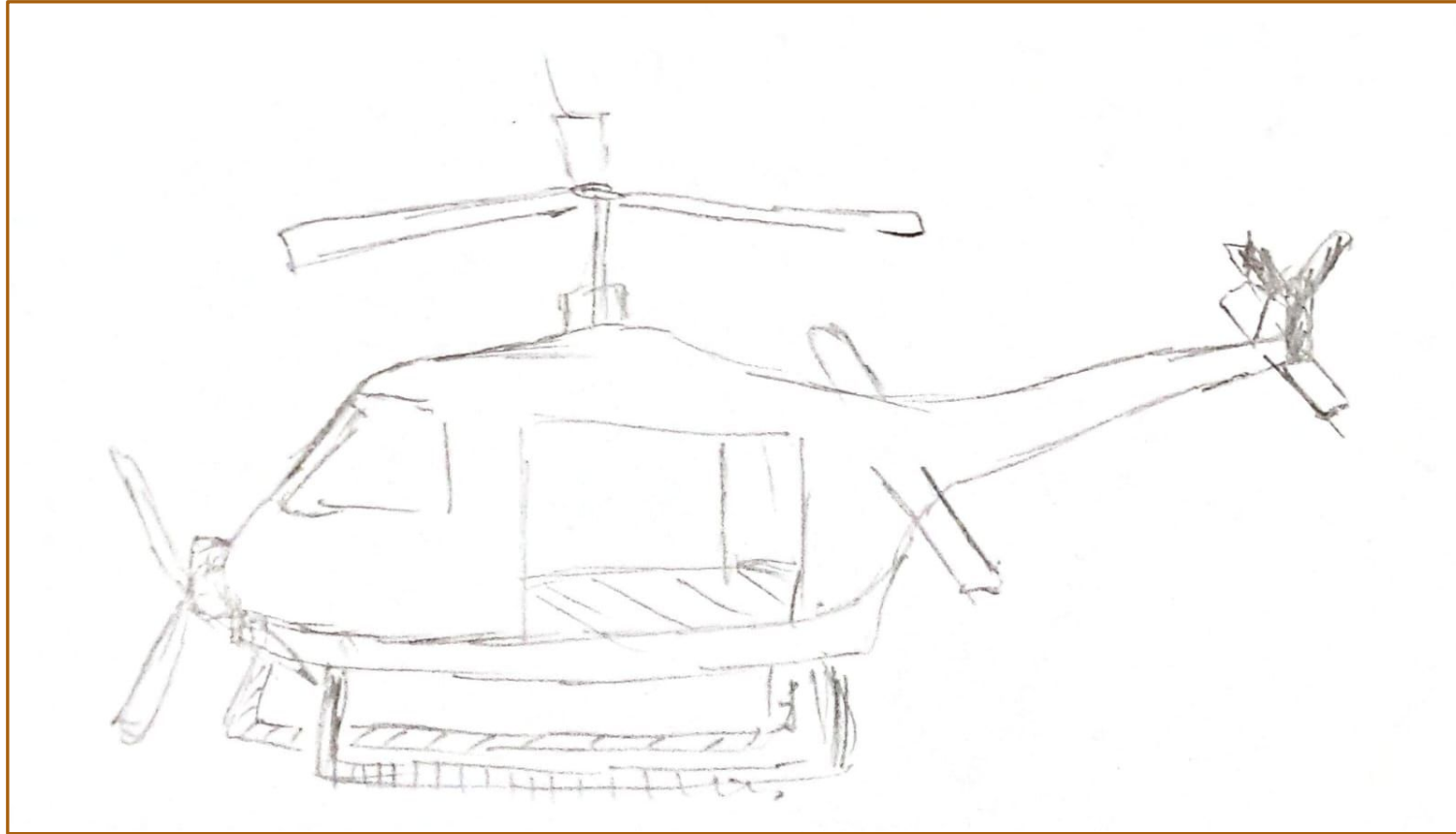
Tail Rotor System

- **Description:** 4-bladed tail rotor
- **Airfoil:** NACA0012
- **Rotor Radius:** 1.1 m
- **Rotor Speed:** 1600 RPM

Powerplant

- **Engine Power:** 1250 kW (upgraded)
- **Fuel Efficiency (SFC):** 0.29 kg/kWh

1.2: Rough schematic sketch of own compound helicopter



Section 2:

Updated Helicopter

Design

0.0°
(Linear)

2.1: Design Parameters of your updated Design

Parameter	Rotor 1	Rotor 2	Rotor ...
Rotor Description (role)	Lifting Rotor	Anti-Torque & Yaw Control	
Airfoil	SC1095	NACA0012	
Rotor Radius (m)	6.6m	1.1m	
Rotor Speed (m)	320	1600	
Number of Blades	4	4	
Chord Length Variation	0.47 m (Constant)	0.18 m (Constant)	
Twist Variation	-10.0° (Linear)	0.0 degrees (Linear)	
Root Cutout	20% of Radius	15% of Radius	

Section 3:

Trim Settings

3.1 Trim settings and resultant forces and moment for 200km/h level flight at 2000 AMSL

Input			Resultant		
theta_0	=	16.82 deg	Fx (Vehicle)	=	-1127.99 N
theta_1s	=	-12.43 deg	Fy (Vehicle)	=	-3247.30 N
theta_1c	=	6.86 deg	Fz (Vehicle)	=	107.33 N
theta_0_t	=	230.52 deg	Mx (Vehicle)	=	-163132.42 Nm
alpha_tpp	=	3.24 deg	My (Vehicle)	=	-51.88 Nm
beta_0	=	22.37 deg	Mz (Vehicle)	=	190.99 Nm

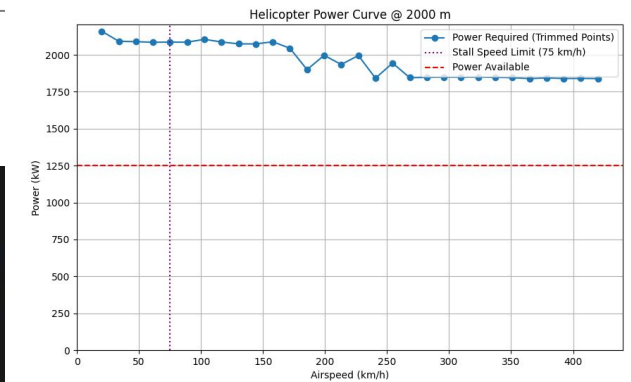
Section 4:

Forward Flight Mission

Test

Findings for your compound helicopter design at 2000 m AMSL

+-----+	
Metric	Value
+-----+	
4.1 Max Speed (Blade Stall)	75.17 km/h
4.2 Max Speed (Power Requirement)	20.00 km/h
4.3 Maximum Range	99.39 km
4.4 Maximum Endurance	1.32 hours
+-----+	



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